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Exercise Sheet 5

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Prove that the set of all irrational numbers is uncountable.

Solution 1.
Write $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$, where $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ is the set of irrationals. We know that $\mathbb{Q}$ is countable and that $\mathbb{R}$ is uncountable.

Suppose, for a contradiction, that $\mathbb{I}$ were countable. The union of two countable sets is countable, so $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$ would be countable, a contradiction. Hence $\mathbb{I}$ is uncountable.

Exercise 2.
Prove or disprove: The set $\mathbb{Z} \times \mathbb{Q}$ is countable.

Solution 2.
The statement is **true**. We have $\mathbb{Z} \preceq \mathbb{N}$ and $\mathbb{Q} \preceq \mathbb{N}$. By Corollary 3.20 we get

$$\mathbb{Z} \times \mathbb{Q} \ \preceq \ \mathbb{N}$$

Hence $\mathbb{Z} \times \mathbb{Q}$ is countable.

Exercise 3.
Prove that the function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined as

$$f(m, n) = 2^{m-1}(2n - 1)$$

is bijective.

Solution 3.
Here $\mathbb{N} = \{1, 2, 3, \dots\}$, so that $m - 1 \geq 0$ and $2n - 1 \geq 1$ and indeed $f(m, n) \in \mathbb{N}$. The idea is that every $k \geq 1$ has a unique decomposition $k = 2^s \cdot q$ with $s \geq 0$ and $q \geq 1$ odd. This follows from the unique prime factorization of k.

Injectivity: Suppose $2^{m-1}(2n-1) = 2^{m'-1}(2n'-1)$, for some $m, n, m', n' \in \mathbb{N}$. Since $2n - 1$ is odd, the highest power of 2 dividing the left side is 2^{m-1}; similarly on the right it is $2^{m'-1}$. From above we know that this decomposition $k = 2^s \cdot q$ is unique, so $m - 1 = m' - 1$, which implies $m = m'$. Since $2^{m-1} \neq 0$, we can cancel it and get $2n - 1 = 2n' - 1$, hence $n = n'$.

Surjectivity: Let $k \in \mathbb{N}$. Write $k = 2^s \cdot q$ with $s \geq 0$ and $q \geq 1$ odd. Set $m = s + 1 \geq 1$ and $n = (q + 1)/2$. Then q odd forces $n \in \mathbb{N}$ with $n \geq 1$, and

$$f(m, n) = 2^{(s+1)-1}(2 \cdot \frac{q+1}{2} - 1) = 2^s\, q = k.$$

Thus f is both injective and surjective, hence bijective.

Exercise 4.
Prove that if A and B are finite sets with $|A| = |B|$, then any surjection $f : A \to B$ is also an injection. Show that this is not necessarily true if A and B are not finite.

Solution 4.
Assume $f : A \to B$ is surjective and A, B finite with $|A| = |B|$. For each $b \in B$ let $F_b = f^{-1}(\{b\})$ be its preimage. Surjectivity means $F_b \neq \varnothing$ for every b, so $|F_b| \geq 1$. Since f is a function, an $a \in A$ can only map to one $b \in B$, so the F_b are disjoint. Further, since every $a \in A$ is mapped to some $b \in B$, the union of the sets F_b covers A. So the F_b form a partition of A. Hence

$$|A| = \sum_{b \in B} |F_b| \geq \sum_{b \in B} 1 = |B|.$$

Since $|A| = |B|$, the inequality can only hold if $|F_b| = 1$ for every b, which implies that f is injective.

Take $A = B = \mathbb{N}$ (so $|A| = |B|$), and

$$f(n) = \lfloor n/2 \rfloor.$$

This is surjective because $f(2m) = m$ for every $m \in \mathbb{N}$, but it is not injective because $f(0) = f(1) = 0$. Thus on infinite sets a surjection from A to B with $|A| = |B|$ need not be injective.

Exercise 5.
Show that $|\mathcal{P}(\mathbb{N} \times \mathbb{N})| = |\mathcal{P}(\mathbb{N})|$

Solution 5.
We define an injection $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by $n \mapsto (n, 0)$. Clearly this is injective. Now we define a function $F : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N} \times \mathbb{N})$ by

$$F(A) = \{f(a) \mid a \in A\}$$

Since f is injective, it is clear that F is also injective.

By Theorem 3.19 there is a bijection $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$. Now we define an injection $G : \mathcal{P}(\mathbb{N} \times \mathbb{N}) \to \mathcal{P}(\mathbb{N})$ by

$$G(B) = \{g(b) \mid b \in B\}$$

Since g is a bijection it is also injective, and by similar reasoning, G is injective.

So now we have $\mathcal{P}(\mathbb{N} \times \mathbb{N}) \preceq \mathcal{P}(\mathbb{N})$ and $\mathcal{P}(\mathbb{N}) \preceq \mathcal{P}(\mathbb{N} \times \mathbb{N})$.

By Theorem 3.16, we conclude that $\mathcal{P}(\mathbb{N} \times \mathbb{N}) \sim \mathcal{P}(\mathbb{N})$

★ **Exercise 6.**

Consider the Cantor triadic set C defined as follows[1]

$$C := \{x \in [0,1] \mid d_i(x) \in \{0,2\}, \forall i \in \mathbb{N}\}$$

Now, the Cantor-Lebesgue function F is defined by

$$F : C \to [0,1], \quad F\left(\sum_{i=1}^{\infty} \frac{a_i}{3^i}\right) := \sum_{i=1}^{\infty} \frac{a_i}{2^{i+1}}$$

(a) Show that $F(0) = 0$ and $F(1) = 1$.

(b) Show that F is well-defined and continuous on C.

(c) Show that F is surjective.

(d) Conclude that C is uncountable.

Solution 6.

a)

Using the base-3 expansions of 0 and $1 = 0.2222\ldots$, we get

$$F(0) = \sum_{i=1}^{\infty} \frac{a_i}{2^{i+1}} = \sum_{i=1}^{\infty} \frac{0}{2^{i+1}} = 0$$
$$F(1) = \sum_{i=1}^{\infty} \frac{a_i}{2^{i+1}} = \sum_{i=1}^{\infty} \frac{2}{2^{i+1}} = \sum_{i=1}^{\infty} \frac{1}{2^i} = 1$$

b)

Well-definedness

Since every number has a unique representation using only $a_i \in \{0,2\}$ and for all $x \in C$ we have

$$0 \leq F(x) \leq \sum_{i=1}^{\infty} \frac{2}{2^{i+1}} = 1$$

F is well defined.

Continuity

Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \varepsilon$. Suppose that $x, y \in C$ with

$$|x - y| < \frac{1}{2^{N-1}} = \delta$$

Assume that x, y match in their first N digits, else $|x - y| < \frac{1}{2^{N-1}}$ would not be possible.

[1] Taken from Prof. Da Lio's Analysis III, Sheet 5

Then cancelling the first N terms gives

$$\begin{aligned} |F(x) - F(y)| &= |\sum_{i=N+1}^{\infty} \frac{x_i - y_i}{2^{i+1}}| \\ &\leq |\sum_{i=N+1}^{\infty} \frac{2}{2^{i+1}}| \\ &= \sum_{i=N+1}^{\infty} \frac{1}{2^i} = \frac{1}{2^N} < \varepsilon \end{aligned}$$

So choosing $\delta = 1/2^{N-1}$ gives

$$|x - y| < \delta \implies |F(x) - F(y)| < \varepsilon$$

Hence F is continuous on C.[2]

c)

Let $y \in [0,1]$ and write $y = \sum_{i\geq 1} b_i/2^{i+1}$ with $b_i \in \{0,1\}$. Then $a_i := 2b_i \in \{0,2\}$ define $x = \sum a_i/3^i \in C$, and $F(x) = \sum a_i/2^{i+1} = \sum b_i/2^{i+1} = y$. Hence F is surjective.

d)

If C were countable, then its image $F(C)$ would be countable (the image of a countable set under any function is countable). But by (c) $F(C) = [0,1]$, which is uncountable..

★ **Exercise 7.**

Prove Theorem 3.16 from the lecture notes.

Theorem (Cantor-Bernstein-Schröder)**.** *Let A and B be sets.*

$$A \preceq B \ \wedge \ B \preceq A \implies A \sim B$$

[2]This is called an $\varepsilon - \delta$ proof. You will see this later in Analysis I.

Solution 7.
$A \preceq B$ gives an injection $f : A \to B$; $B \preceq A$ gives an injection $g : B \to A$. Then notice that $g : B \to g(B)$ is a bijection from B to the range of g.[3]. Since it is bijective it has an inverse $g^{-1} : g(B) \to B$. Now consider the subset $G \subseteq A$ defined as follows

$$G = \bigcup_{k=0}^{\infty} (g \circ f)^k (A \setminus g(B))$$

Define $W = A \setminus G$, so $A = G \cup W$. Now we define a function $h : A \to B$ as

$$h(x) = \begin{cases} f(x) & \text{if } x \in G \\ g^{-1}(x) & \text{if } x \in W \end{cases}$$

If $x \in W$ then $x \notin G$. This implies that $x \notin A \setminus g(B)$. So we have that $x \in g(B)$ and hence $g^{-1}(x)$ is well defined in this case.

Now it remains to show that h is injective and surjective.

Let $x, y \in A$ such that $h(x) = h(y)$. We proceed by cases.

Case 1: $x, y \in G$. Then by definition $h(x) = h(y)$ implies that $f(x) = f(y)$. Since f is injective, we get $x = y$

Case 2: $x, y \in W$. Then by definition of h we get that $g^{-1}(x) = g^{-1}(y)$. Since g is well-defined we can apply it again and get $x = y$.

Case 3: x, y are in different sets. Without loss of generality assume that $x \in G$ and $y \in W$. By the definition of G we have $x = (g \circ f)^k(z)$ for some $k \geq 0$ and $z \in A \setminus g(B)$. Now notice that $h(x) = h(y)$ implies that $f(x) = g^{-1}(y)$. This means that $f((g \circ f)^k(z)) = g^{-1}(y)$. Since g is well-defined we can apply it to both sides and get $(g \circ f)^{k+1}(z)$. This implies that $y \in G$. However this is a contradiction since $y \in W$. So the third case can never happen.

Since the cases clearly cover all possibilites, h is injective.

Let $b \in B$ be arbitrary. Notice that $g(b) \in A$ so either $g(b) \in W$ or $g(b) \in G$. In the first case, we have that $h(g(b)) = g^{-1}(g(b)) = b$. So taking $x = g(b)$ fulfills the property of surjectivity.

Now assume that $g(b) \in G$. By the definition of G we have $g(b) = (g \circ f)^k(z)$ for some $k \geq 0$ and $z \in A \setminus g(B)$. Now assume for contradiction that $k = 0$. Then $g(b) = (g \circ f)^0(z) = z \in A \setminus g(B)$. But clearly $g(b) \notin A \setminus g(B)$, so we get that $k > 0$. By definition of function composition we get

$$\begin{aligned} g(b) &= (g \circ f) \circ (g \circ f)^{k-1}(z) \\ &= g\left(f((g \circ f)^{k-1}(z))\right) \end{aligned}$$

Since g is injective this implies that $b = f((g \circ f)^{k-1}(z))$. Let $x = (g \circ f)^{k-1}(z)$, so $x \in G$ by definition. Now observe that $h(x) = f(x) = f((g \circ f)^{k-1}(z)) = b$. Thus h is surjective.

Since h is both injective and surjective, it is a bijection.[4]

[3]This also called a restriction of g to $g(B)$

[4]This proof is adapted from *The Book of Proof*. If you have trouble visualizing the construction, see Figures 14.3–14.6.